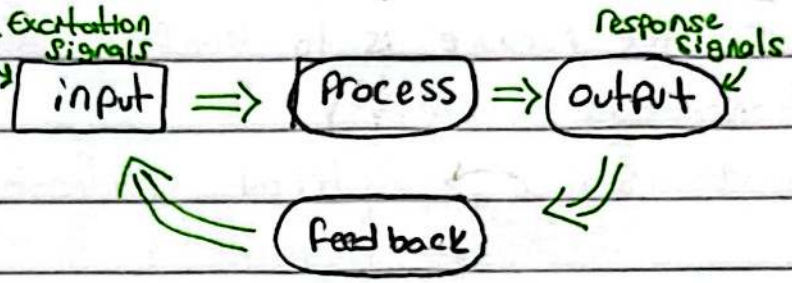
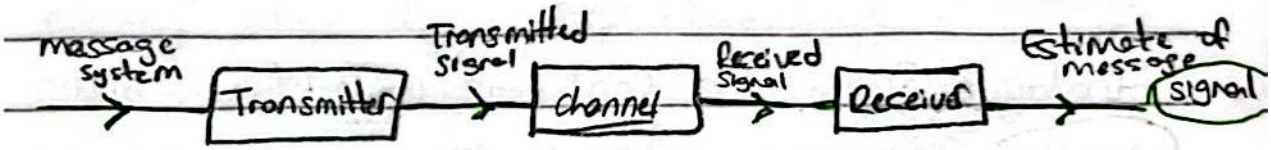
BLM2041 Signals and Systems 23-24

Sena Cindioğlu

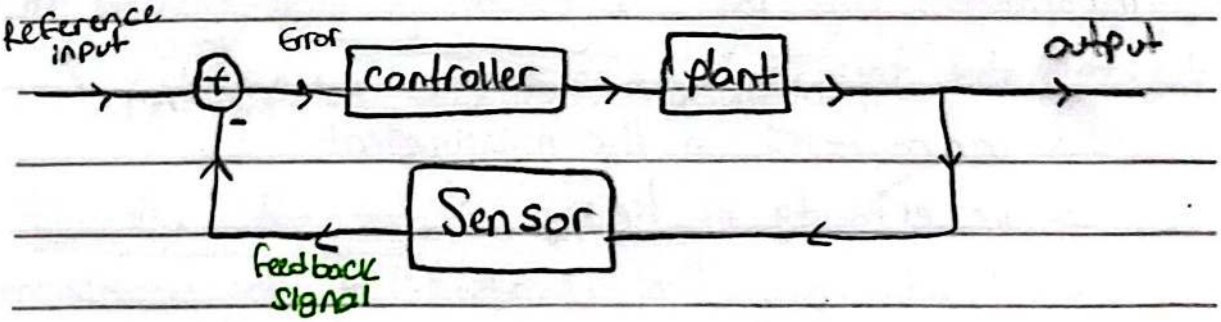
System : integrated set of elements or can be defined as multiple parts working together for a common purpose



→ Communication System:



→ Feedback Control system:



Digital System

discrete inputs → Disc info processing system → Discrete outputs

Signal: a pysical thing that conveys information and varies with an indepent var such as time

⊛ Transducer: is a device that converts energy from one to another. the purpose is to tranform information

① input transducers: (sensors)

non-electrical → electrical (microphone)

② output transducers:

electrical → non-electrical (speaker)

Analogue Signal: a continuos var defined with infinite precision ↑~→

information is lost in converting from analogue to digital, due to:

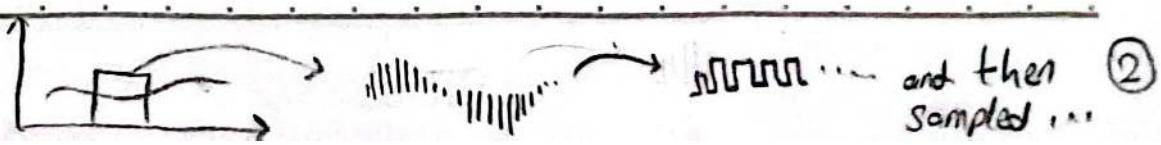
→ Inaccuracies in the measurment

→ uncertainty in timing

→ limits on the duration of the measurement

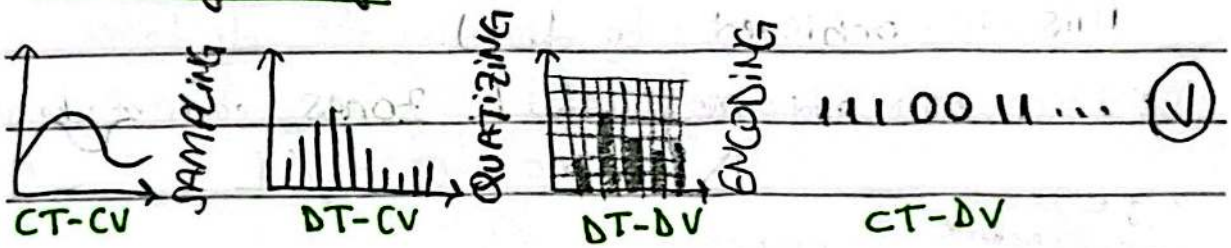
these effects are called quantisation errors

Digital Signal: when the cont. analogue signal held, the measurment would be converted to a digital value. because if it isnt held the signal would be changing



Analog to Digital Conversion (ADC):

1. Filtering : Filter the signal to limit the max frequency
2. Sampling : usually taken at equal intervals of time (eşit aralıklarla)
3. Quantization
4. Binary Coding



(*) Sampling : Analog Signal is sampled every Ts Secs. (Ts aralıkla örnekleme)
Sampling rate = $f_s = 1/T_s$ (frequency)
there are 3 Sampling methods:

1- Ideal : an impulse at each sampling instant
2- Natural : a pulse with a short wide with varying amplitude
3- Flattop : like natural but with single amplitude (genlik)
→ PAM (Pulse amplitude modulation)

Sampling theorem:

According to the Nyquist theorem, the Sampling rate must be at least 2 times the highest frequency contained in the signal $F_s \geq 2 f_{max}$ (önemli)

→ örneklem orjinal veriye dönebiliyorsa, bu örneklem uygundur (enterpolasyon, regresyon ...)

$$V_{LSB} = \frac{V_{max}}{2^{\text{number of bits}}}$$

Quantization:

the amplitude values between the min and the max in the Sampling are infinite → we need to map them onto a finite set of known values

this is achieved by dividing the distance between min and max into L zones each of height Δ

$$\Delta = (max - min)/L$$

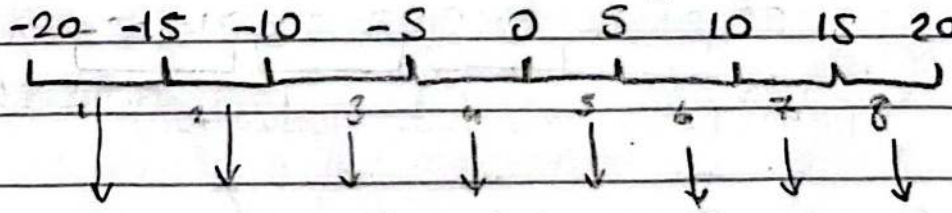
Zones:

assume that $V_{min} = -20$ and $V_{max} = +20$

$L = 8$ → (zones) quantization levels

Δ = zone width = $20 - -20 / 8 = 40/8 = 5$

$\Delta = 5$ So the zones are ↓



the mid points → -17.5, -12.5, -7.5, -2.5, 2.5, 7.5, 12.5, 17.5

Assigning codes to zones → $n_b = \log_2 L$

$8L = 3 = n_b$ → 000, 001, 010, 011, 100, 101, 110, 111

-20,-15 -15,-10 -10,-5 -5,0 0,5 5,10 10,15 15,20

* the more zones → the smaller Δ → smaller Errors. But the more bits required to encode

Example:

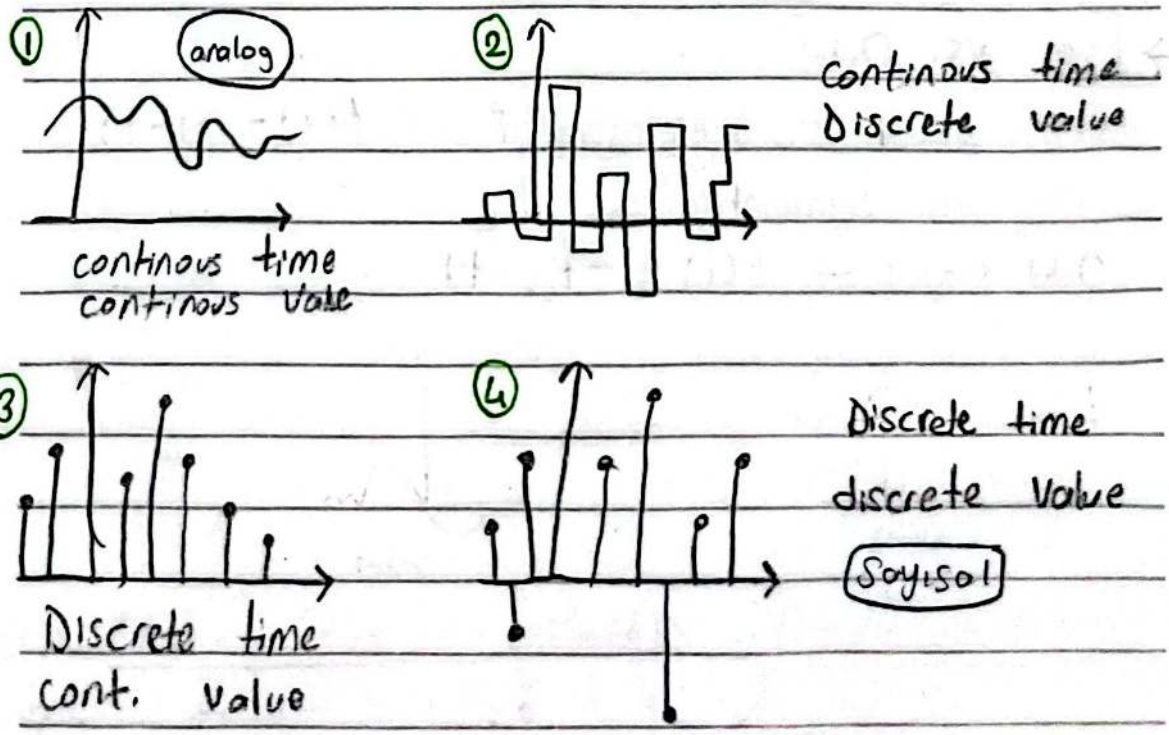
an 12 bit ADC advertises an accuracy of ~~F the~~ least significant bit if the input range of the ADC is 0 to 10 volts, What is the accuracy of the ADC in analoge volts?

$$V_{LSB} = \frac{V_{max}}{2^{\text{No. of bits}}} = \frac{10}{2^{12}} = 10/4096 = 0.0024 \text{ volts}$$

(No. of bits: number of bits)

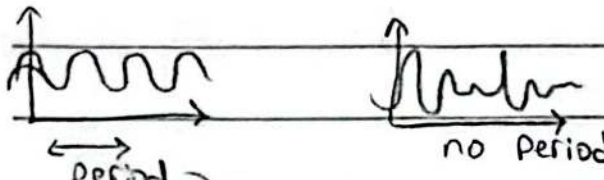
hence the accuracy = ± 0.0024 volts

(X) Types of signals



Classifications of Signals:

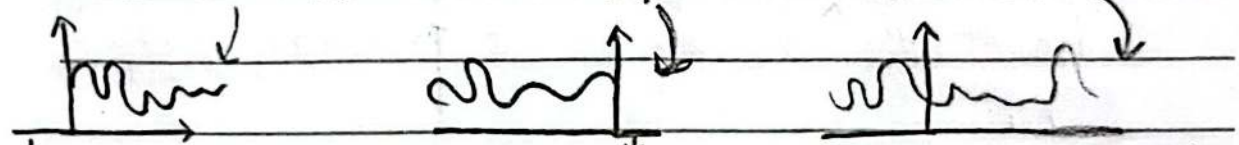
→ Periodic vs. Aperiodic



$f(t) = f(t+T)$

⊛ Fakını anlamamız için frekans değerine sonra period değerine bakarız tüm değerler aynıysa periodic olur

Causal vs. Anticausal vs. Non causal

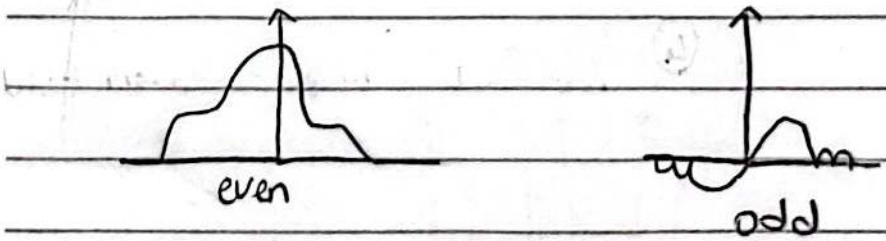


→ Even vs. Odd

even signal is any signal f → $f(t) = f(-t)$

they are symmetric

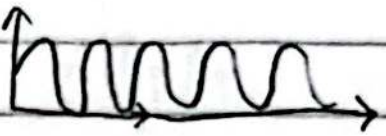
Odd signal → $f(t) = -f(-t)$

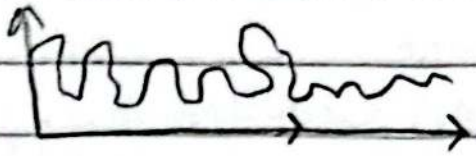


$\frac{x(t) + x(-t)}{2} = x(t)_{even}$ → $x(t) = x(-t)$

$\frac{x(t) - x(-t)}{2} = x(t)_{odd}$ → $x(t) = -x(t)$

→ Determenistic vs. Random

Determenistic signals which each value being determined by rule 

Random 

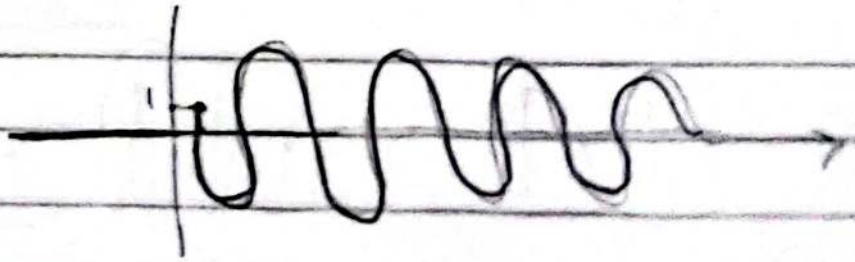
Example

Consider the sigral defined

$$f(t) = \begin{cases} \sin(2\pi t)/t & t \geq 1 \\ 0 & t < 1 \end{cases}$$

contious time, analog, aperiodic, infinite causal, neither even nor odd, and deterministic

Both if

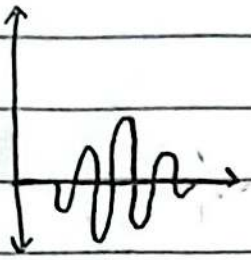


causal

Week -3-

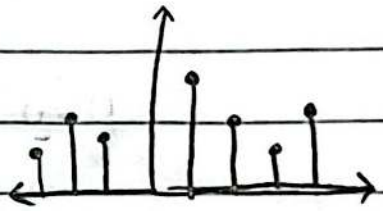
1- Continuos Systems (time) v(t)

- Am or FM radio
- cont inputs and outputs



2- Discrete Systems (time) x[n]

Pc comp. game
disc time input and output
sequence of Samples

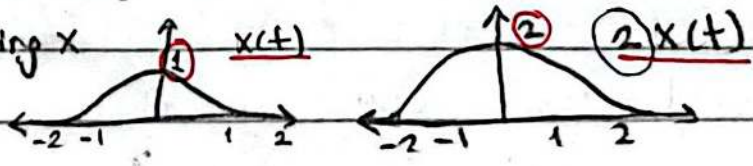


3- Hybrid systems (A/D , D/A converters)

Operations on Systems :

① Amplitude Scaling (genişleme) :

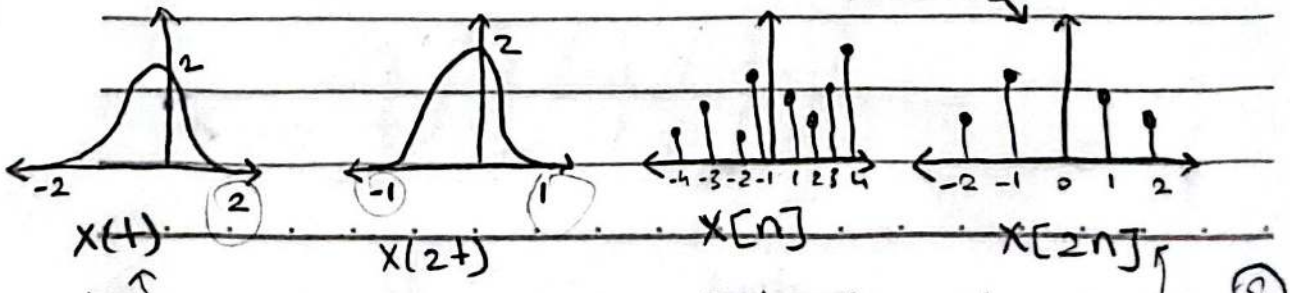
→ multiplying x
y'yi etkiler



② - Time Scaling (daralması) : yada genişleme

→ multiplying the 't' or the 'n' x'i değiştirir

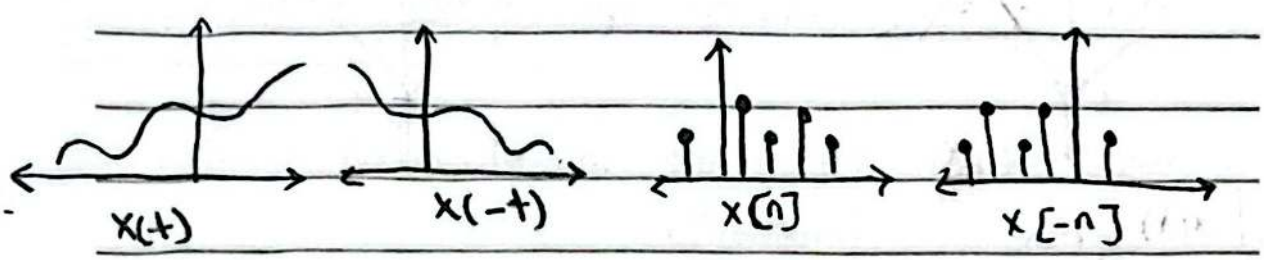
bölseydik daha fazla sıklık olurdu



tam tersi — çarparsak genişler, bölersek daralır

③- Time Reversal: (-1 ile çarpma) y'ye simetri

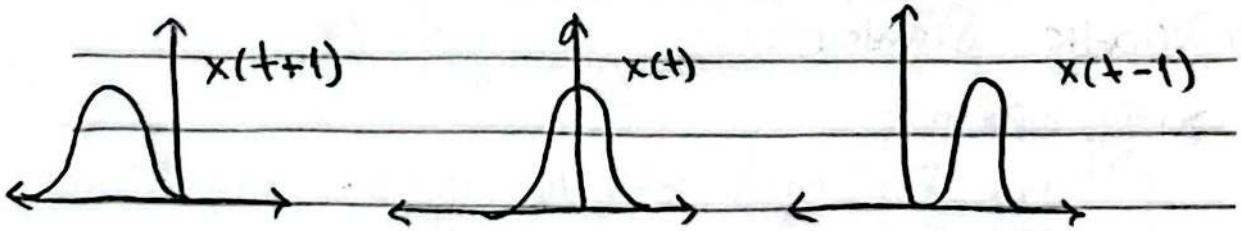
→ replace t with (-t) or n with (-n)



④- Time Shift: (kaydırma, ötelemе) (x ekseni kaydır)

→ replace x(t) with x(t-m) or x(t+m)

Same for x[n] → x[n-m] or x[n+m]

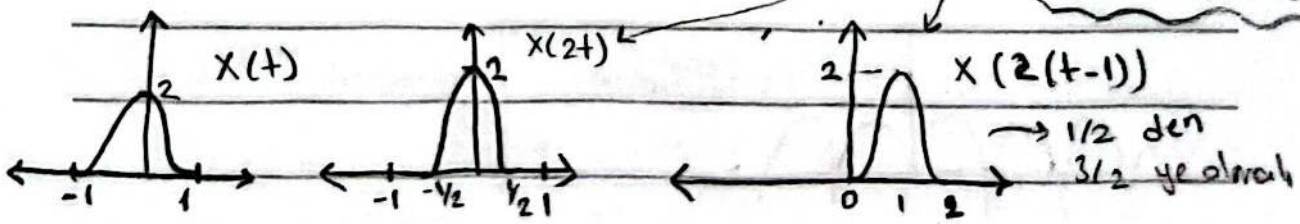


→ (+) olursa sola, (-) olursa sağa.

✳ Combination Operations:

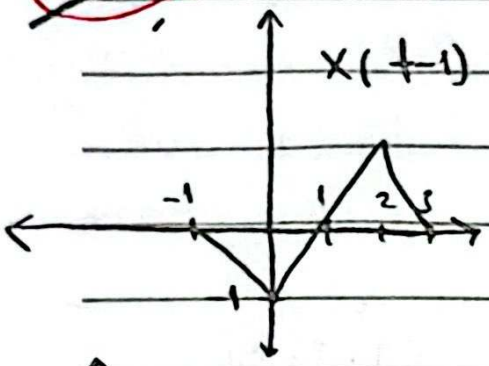
→ Scale first then shift example x(2(t-1)) tam tersi

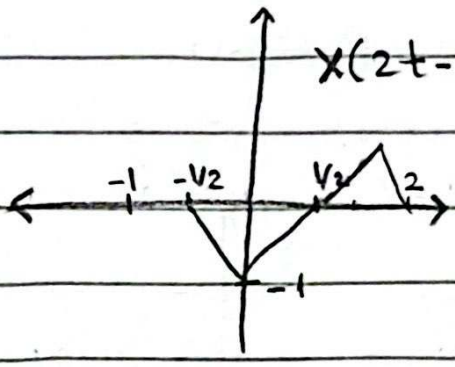
ilk çarpma/bölme sonra toplama/çıkma

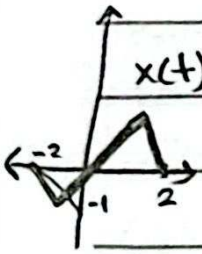


→ 1/2 den 3/2 ye alıralım

Soru







$y(2t-1) = ?$ Önce zamanda ötele

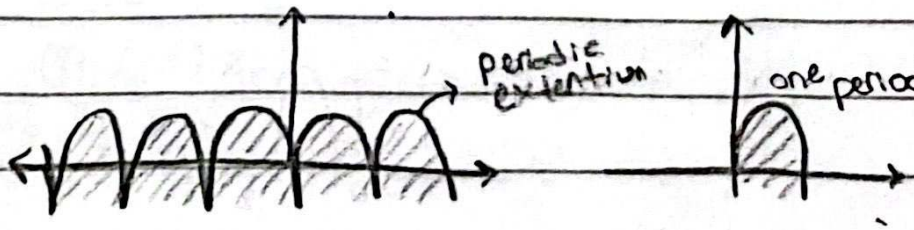
Sonra ölçekle

⊛ Periodic Signals !

→ very important

* cont time signal is periodic $\xrightarrow{\text{if}}$ $T_0 > 0$ such that $x(t+T_0) = x(t)$

* disc. time signal is periodic $\xrightarrow{\text{if}}$ $N_0 > 0$ such that $x[n+N_0] = x[n]$



Yani belirli bir T zamanı sonrası aynı sinyale eşit oluyorsa $x(t+T) = x(t)$

→ Complex Signals:

$z(t) = x(t) + jy(t)$ where $j = \sqrt{-1}$

- Arises naturally in sinusoids, Radar, Sonar... ultrasound and many communication

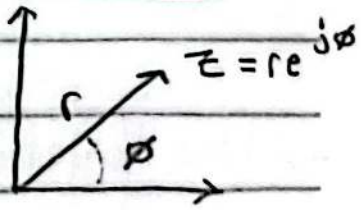
- Review -

$z = x + jy$ → complex number

- x: the real part
- y: the imaginary part
- engineering notation of $i = \sqrt{-1}$

→ Complex number in Polar form: $z = re^{j\phi}$

- $r = z$
- ϕ the angle of phase z
- $\exp(j\phi) = \cos\phi + \sin\phi$



$$e^z = e^{x+jy} = e^x e^{jy} = e^x(\cos y + j\sin y)$$

$$e^{j\pi\alpha} = \cos\alpha\pi + j\sin(\alpha\pi)$$

! Çok kullanılacak türetmeleri, çevirmeyi iyi bil

$$\cos\alpha\pi = \frac{e^{j\pi\alpha} + e^{-j\pi\alpha}}{2}$$

$$\sin\alpha\pi = \frac{e^{j\alpha\pi} - e^{-j\pi\alpha}}{2j}$$

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Week -4-

Signal Energy and Power

- if i(t) is the current through a resistor, then the energy dissipated in is :

$$E_R = \lim_{T\to\infty} \int_{-T}^{T} i^2(t) R \, dt \quad \text{in Joules}$$

- The Signal Energy for i(t) is defined as the energy dissipated in a 1 Ω resistor:

$$E_i = \lim_{T\to\infty} \int_{-T}^{T} i^2(t) \, dt$$

- The Signal energy for (complex) signal is

$$E_x = \lim \int |x(t)|^2 \, dt$$

→ The average of the signal energy over time is the signal power

⊛ An energy signal x(t) has Zero Power

$$P_x = 0 \rightarrow \lim 1/2T \int |x(t)|^2 \, dt$$

⊛ A power signal has infinite energy

$$E_x = \infty \rightarrow \lim 2T \frac{1}{2T} \int |x(t)|^2 \, dt$$

$$P = \frac{E_0}{T_0}$$

- Sinusoidal Signal - Önemli

$$x(t) = \cos(\omega t + \theta)$$

(frequency, faz (fi))

$$\omega = 2\pi f = 2\pi / T$$

or

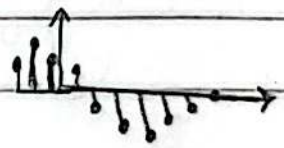
$$x(t) = A\cos(2\pi f t + \theta)$$

ω ; radian frequency f ; frequency in herts

the period is $T = \frac{1}{f} = \frac{2\pi}{\omega}$

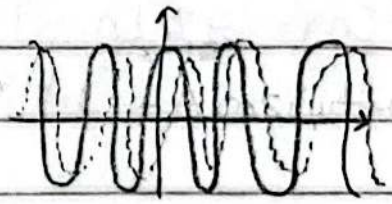
in its discrete time form:

$$A\cos(\omega n + \varphi)$$



Complex Sinusoids

$$e^{j\theta} = \cos\theta + j\sin\theta$$

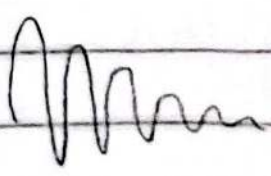


$$Ae^{j(\omega t+\theta)} = \underbrace{A\cos(\omega t+\theta)}_{\text{Real sinusoid}} + jA\sin(\omega t+\theta)$$

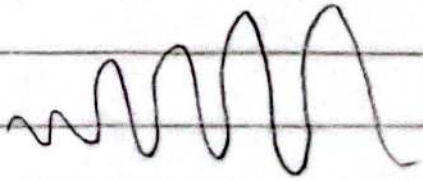
like $x + jy$

Exponential Signals

$$x(t) = e^{\alpha t} \qquad \alpha<0 \downarrow \quad \alpha>0 \uparrow$$



$e^{\alpha t}\cos(\omega t)$

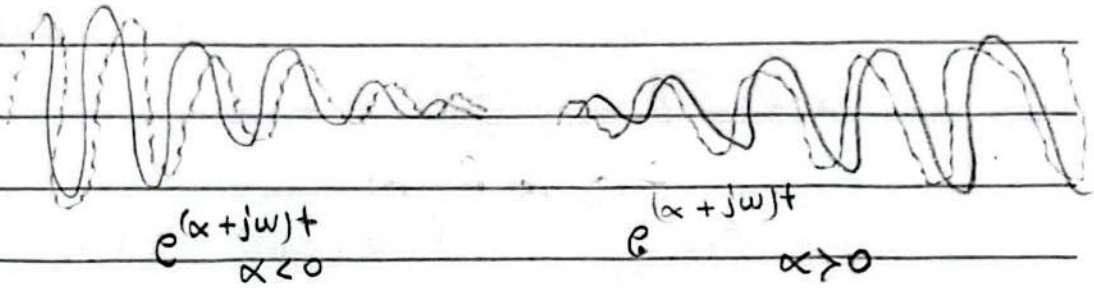


$e^{\alpha t}\cos(\omega t)$

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Complex exponential Signals

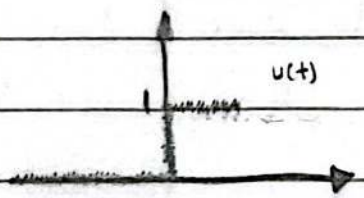
$$e^{(\alpha + j\omega)t + j\theta} = e^{\alpha t}(\cos(\omega t + \theta) + j\sin(\omega t + \theta))$$



Unit Step Functions Önemli

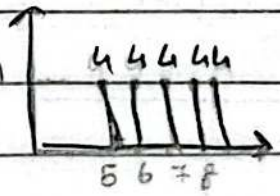
Heaviside step

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

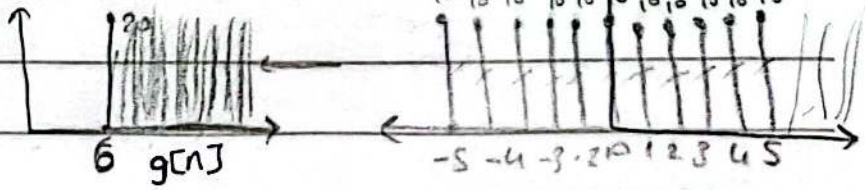


Örnek

→ $f[n] = 4u[n-5]$



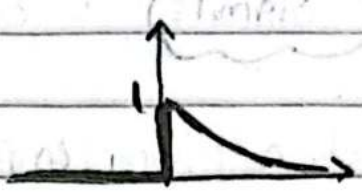
→ $g[n] = 10u[n-5] + 10u[n+5]$



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Uses for unit step

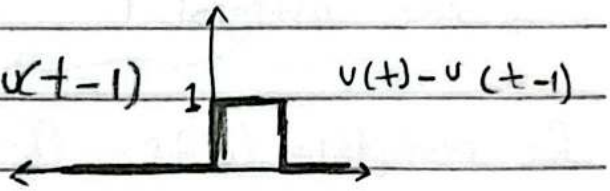
① $x(t) = u(t)\, e^{-t}$



② the offset rectangular Signal

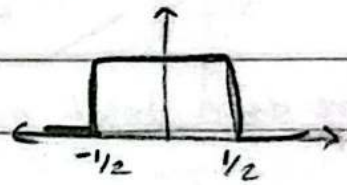
$x(t) = u(t) - u(t-1)$

$$x(t) = \begin{cases} 0, & t \geq 1 \\ 1, & 0 \leq t < 1 \\ 0, & t < 0 \end{cases}$$



③ Unit Rectangle

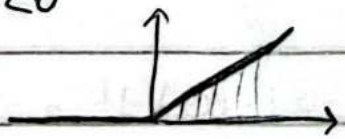
$$rec(t) = \begin{cases} 1 & |t| \leq 1/2 \\ 0 & \text{otherwise} \end{cases}$$



④ Unit Ramp

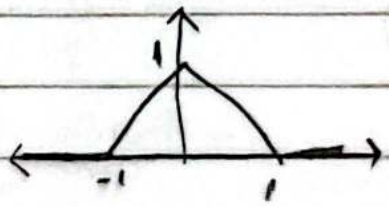
$$r(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

integral → $r(t) = \int_{-\infty}^{t} u(\tau)\, d\tau$



⑤ Unit triangle

$$\Delta(t) = \begin{cases} 1 - |t| & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

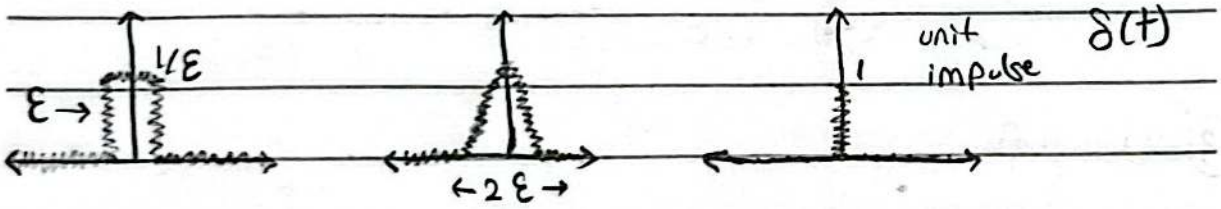


Impulsive Signals

önemli:

delta function or impulse (δ) [Dirac]
is an idealization of a signal that;
→ very large near $t=0$
→ very small away from $t=0$
→ has integral 1

for example:



the shape doesn't matter (ε is small)

✱ Properties:

① $\int_{-\infty}^{\infty} f(t)\cdot\delta(t)\,dt = f(0)$ $\left(f(0)\cdot 1 = f(0)\right)$ ($\delta(0)$) $f(t)\approx f(0)$

② $\int_{-\infty}^{\infty} \alpha\cdot\delta(t-\theta)\cdot f(t)\,dt = \alpha\cdot f(\theta)$ $(t=\theta$ için $\delta(0)=1)$

③ Sampling: $\phi(t)\,\delta(t)$?? $\int f(t)\phi(t)\delta(t) = f(0)\cdot\phi(0)$ ← if you find this you find the sample

④ Limits of integration
+ if $a<0$, $b>0$ $\int_a^b \delta(t)\,dt = 1$ → if $a>0$, $b<0$ $\int_a^b \delta(t)\,dt = 0$

$a=0$, $b=0$ } belirsiz

Example:

$$\int_{-2}^{3} f(t)\,(2+\delta(t+1)-3\delta(t-1)+2\delta(t+3))\,dt$$

$$2\int_{-2}^{3} f(t)\,dt + \int_{-2}^{3} f(t)\cdot\delta(t+1)\,dt - 3\int_{-2}^{3} f(t)\cdot\delta(t-1)\,dt$$

$$+2\int_{-2}^{3} f(t)\,\delta(t+3)\,dt =$$

$f(-1)\cdot 1$; $f(1)\cdot 1$

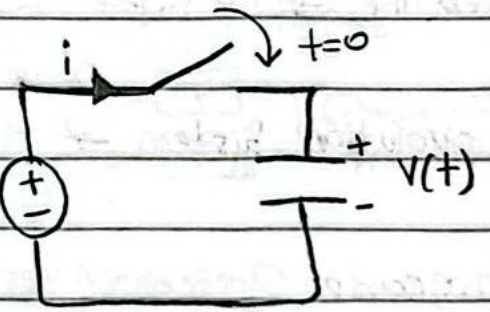
$f(-3)\cdot 1 = 0$ $\boxed{-3<-2}$

$$\boxed{2\int_{-2}^{3} f(t)\,dt + f(-1) - 3f(1) + 0}$$

Example:

assuming $v(0)=0$

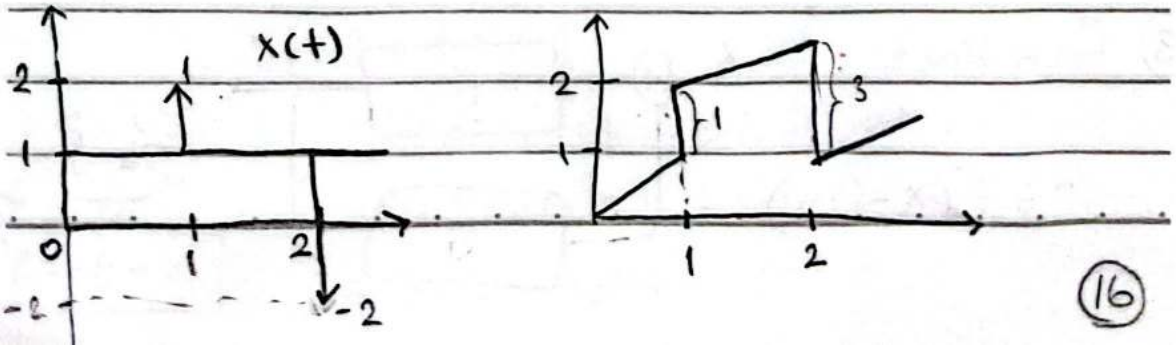
what is $v(t)$, $i(t)$ for $t\geq 0$



- $i(t)$ is very large for very short moment
- unit charge → capacitor
- $v(t)$ increases to $v(t)=1$ (anında, Gok kısa zamanda)

Example:

$x(t) = 1+\delta(t-1)-2\delta(t-2)$ define $y(t) = \int_{0}^{t} x(\tau)\,d\tau$



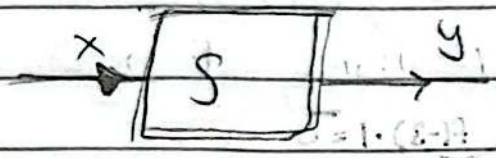
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Son Gözdük

Week -5-

Systems $y = S(x)$

we will concentrate on systems with one input and one output (single input single output) SISO



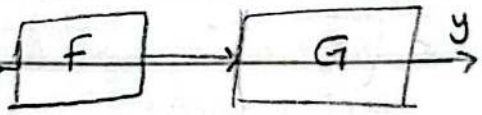
Scaling System $y(t) = a\,x(t)$

- amplifier → $|a| > 1$
- attenuator → $|a| < 1$
- inverting → $a < 0$

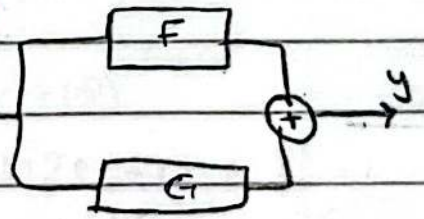
a is called scale factor (the gain)

* convolution System → $y(t) = \int x(t-\tau)h(\tau)\,d\tau$

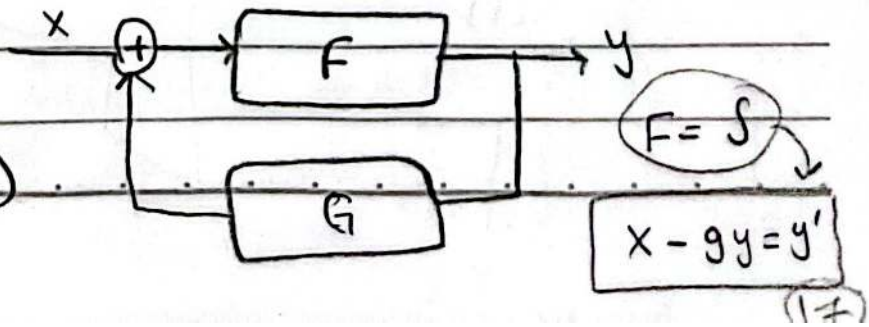
① Cascade System (seri): $G(F(x)) = y$



② Sum System (Paralel): $y = Fx + Gx$



③ Feed Back: $y = F(x - Gy)$

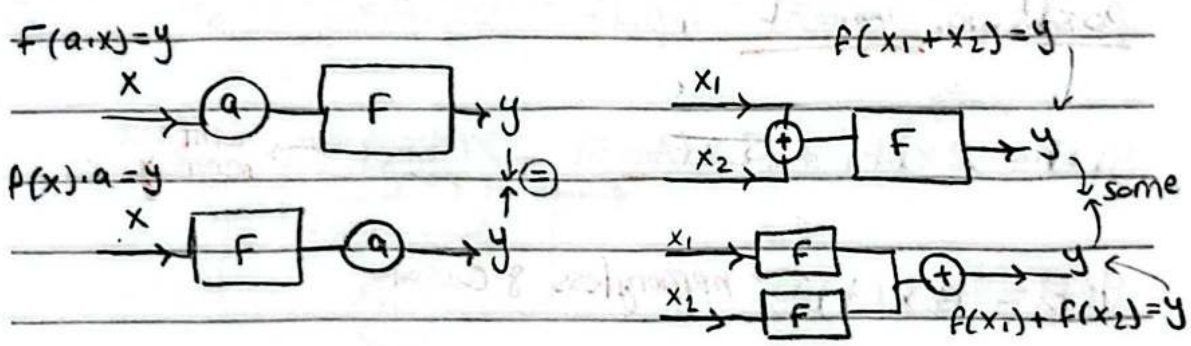


⊗ Linearity → toplomsallık ve carpomsallık şartları gerçekleşirse sistem lineer dur

→ homogeneity: $F(ax) = a\,F(x)$ ⊗ f(6)=2 f(3)

→ Super Position: $F(x+y) = F(x) + F(y)$ ⊗ f(5) = f(2) f(3)

• in short scaling and summing before and after the System is Same



✗ $y(t) = x^2(t)$ nonlineer ✗ ✓ $x(t)\cdot\cos(2\pi ft)$ lineer ✓✓

Örnek

$y(t) = t\,x(t)$ = ?? lineer mi

$x_3(t) = a\,x_1(t) + b\,x_2(t)$

$x_1(t) \to y_1(t) = t\,x_1(t)$

$x_2(t) = y_2(t) = t\,x_2(t)$

$y_3(t) = t\,x_3(t)$

$= t(a\,x_1(t) + b\,x_2(t))$

$= a\,t\,x_1(t) + b\,t\,x_2(t) =$

$= y_1(t) + y_2(t)$ ✓✓

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System Memory:

1- memoryless System: if the output depends only on the present input (amplifier)

2- with memory System: if the output depends on inputs on the past or future (capacitor)

3- Causal System: if the output depends only on past or present inputs (past, present)

$y(t) = 2x(t) + 5x(t-3) - x(t+2)$ → with memory

$y(t) = 6x(t)$ → memoryless & causal

$y[n] = 3x[n] - x[n-7]$ → causal

Time Invariance: (Zamanda degişkenlik)

$y(t) = Fx(t) \rightarrow y(t-\emptyset) = Fx(t-\emptyset)$ for any $\emptyset$

System Stability:

if $|x(t)| \leq M_x < \infty$ then $|y(t)| \leq M_y < \infty$

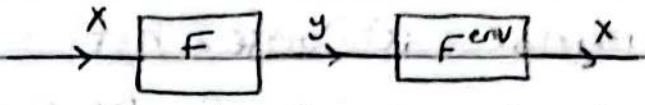
the system is bounded input bounded output → stable (BIBO)

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System Invertibility

$y = Fx$

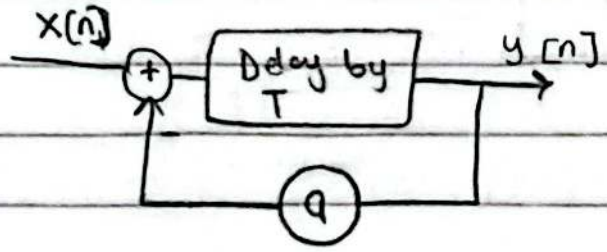
$x = F^{inv}_{y} = F^{inv}F(x)$



Discrete - Time Systems

→ equations are difference equations :

$$a_0 y[n] + a_1 y[n-1] + \cdots = b_0 x[n] + b_1 x[n-1] \cdots$$



- $e[n] = x[n] - ay[n]$ ↘

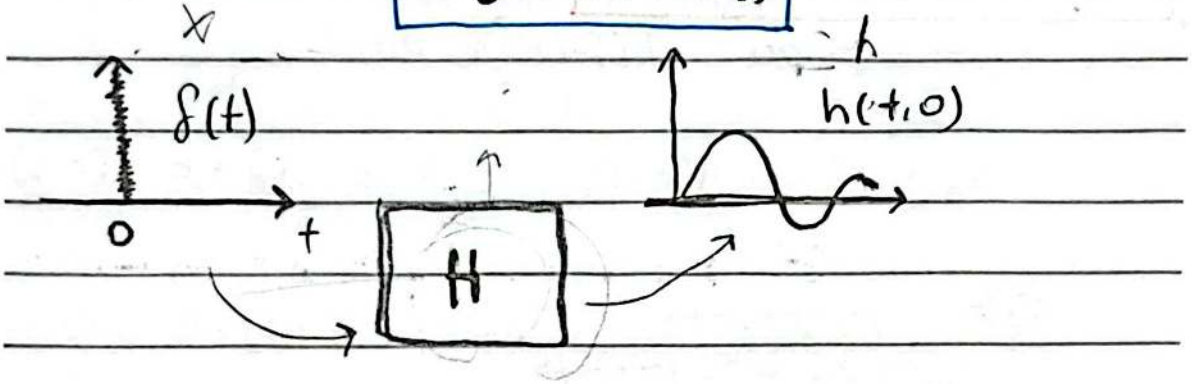
- $y[n] = e[n-1]$ so $y[n] = x[n-1] - ay[n-1]$

Week -6-

Convolution en önemli konu

⊛ impulse response: $h_\tau(t)$
is the output of the system at time t to an impulse at time τ

$$h_\tau = H(\delta_\tau)$$



⊛ $h(2) \neq H(\delta(2))$ → $(\delta(2) \approx 0 \rightarrow h(2) \neq H(0))$

→ **Linearity:**
A system S is linear if it both: ① homogen + ② superposition

① $y = Sx \rightarrow ay = S(ax)$: a = constant

② $y_1 + y_2 = S(x_1 + x_2)$

• Liner system : $ay_1 + by_2 = S(ax_1 + bx_2)$ a and b costat

• Extended Linearity : doğru ve güvenilir performansın geniş koşullar altında sağlanmasını amaçlar. (21)

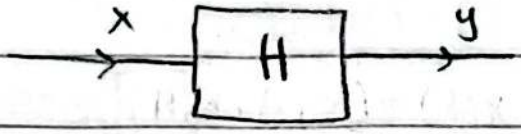
⊙ Summation
$\sum a_n y_n = S(\sum a_n y_n)$

⊙ integration
$\int a(\tau)\, y(t-\tau) = S\left[\int a(\tau)\, x(t-\tau)\right]$

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⊛ Linear Time-Invariant System: (LTI)

we can write a signal $x(t)$ as sample of itself



$$x(t) = \int x(\tau)\,\delta_\tau(t)\,d\tau$$

$$y(t) = H\left(\int x(\tau)\,\delta_\tau(t)\,d\tau\right) = H(x(t))$$

and if H is time invariant, this written more simply as (LTi)

$$y(t) = \int x(\tau)\,h_\tau(t)\,d\tau$$

this the form of a convolution int.

(LTI)

$$x(t) = \int x(\tau)\delta(t-\tau)\,d\tau \xrightarrow{\text{system}} y(t) = \int x(\tau)h(t-\tau)\,d\tau$$

Convolution: is an integral that expresses the amount of overlap of one function when its shifted over another function.

$$y(t) = h(t) \underset{\text{convolution}}{*} x(t) = \int_{-\infty}^{\infty} x(\tau)\cdot h(t-\tau)\,d\tau$$

(*) $x(t) * A\,\delta(t-h) = A\,x(t-h)$

(*) Convolution Properties

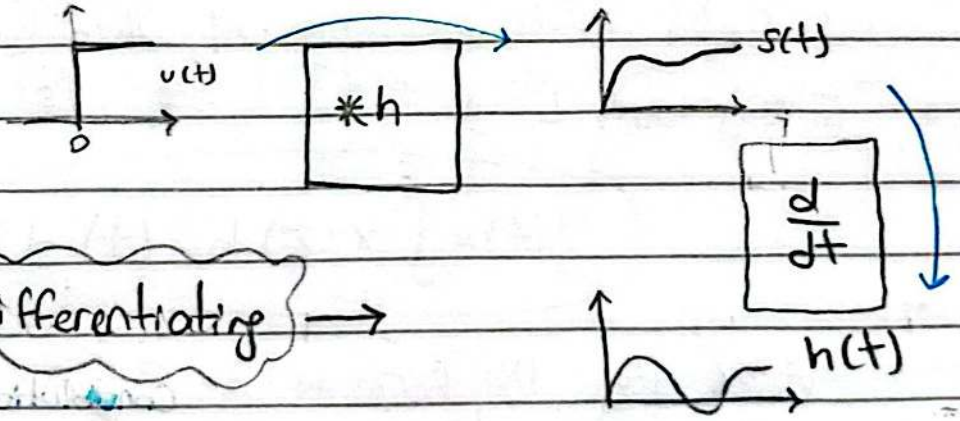
- commutativity → $x(t) * h(t) = h(t) * x(t)$

- Associativity → $[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$

(*) Distributivity → $x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$

- Derivative → $d/dt\,[x(t) * v(t)] = \dot{x}(t) * v(t) = x(t) * \dot{v}(t)$

- A common alternative to measure the step response
$s(t)$ → the response to a unit step input $u(t)$



differentiating + convolution!

$u(t) \rightarrow d/dt \rightarrow \delta(t) \rightarrow *h \rightarrow h(t)$

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Steps :

① $x(t) \rightarrow x(\tau)$, $h(t) \rightarrow h(\tau)$

② $h(-\tau)$ scaling

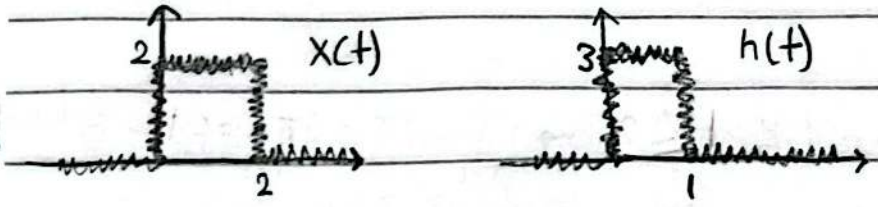
③ $h[-(\tau - t)] = h(t-\tau)$ time shifting

④ $x(\tau) \cdot h(t-\tau)$

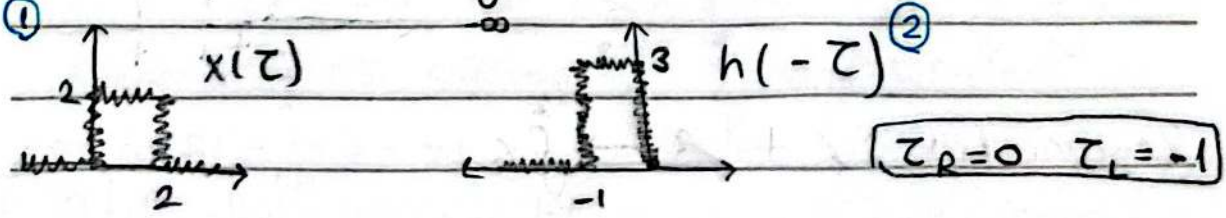
⑤ $\int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$

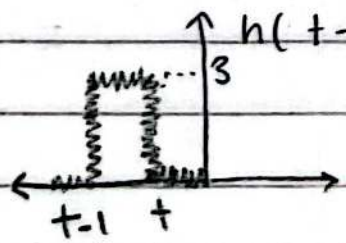
example:

convolve these two signals



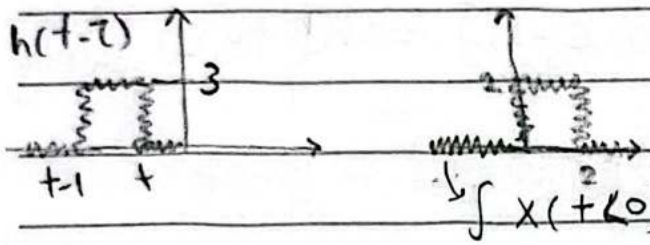
formal $y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$





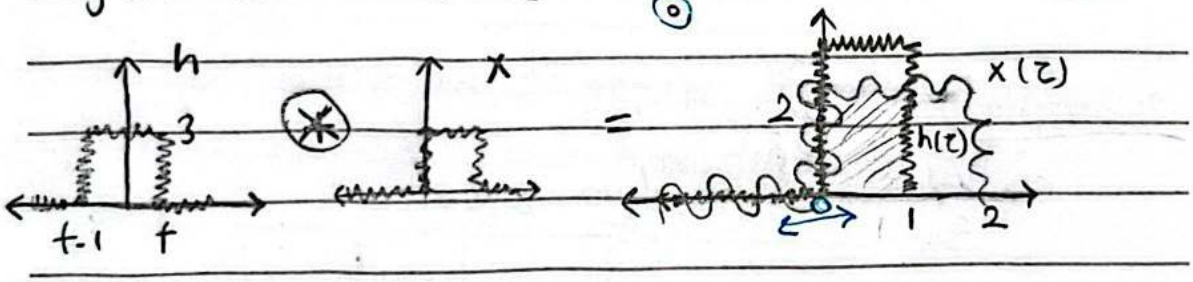
④ Now we must find the regions (integral alanları)

Region 1 : $t<0 \rightarrow \int x(\tau)\cdot h(t-\tau) = \boxed{0}$

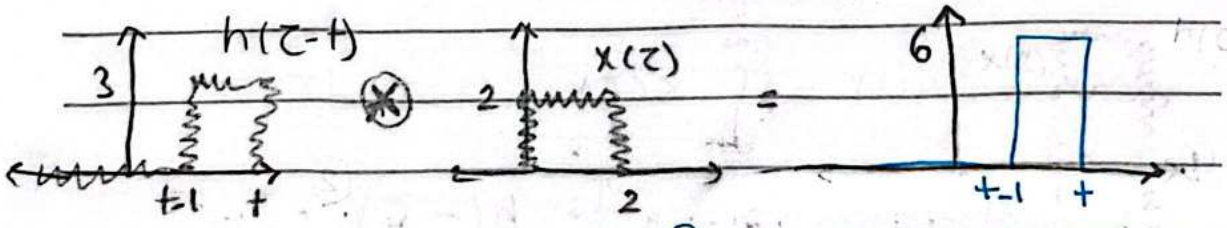


$\int x(t<0) = 0$ always

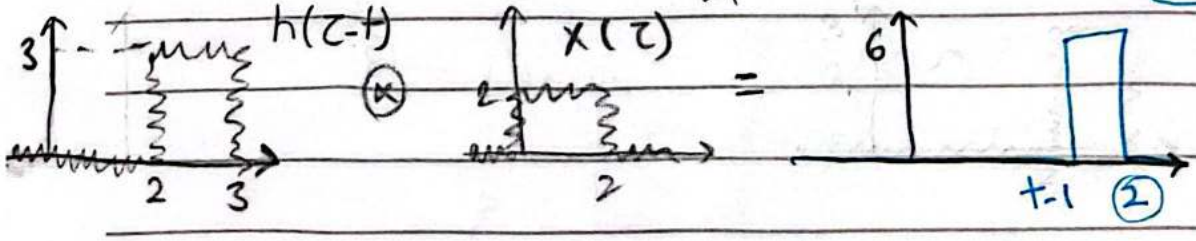
Region 2 : $0 \le t \le 1 \Rightarrow \int_0^t 6\,d\tau = 6t - 6\cdot 0 = \boxed{6t}$



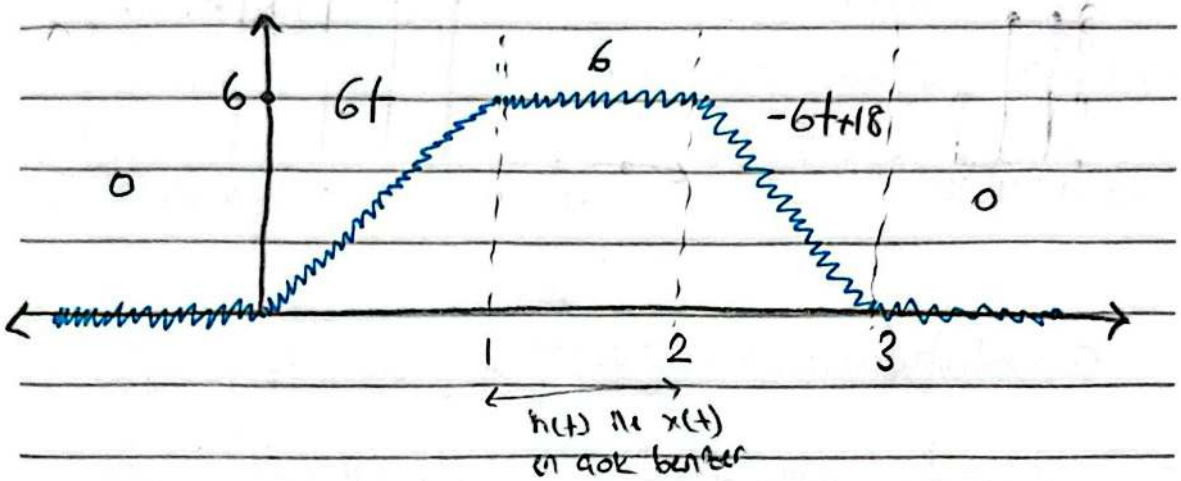
Region 3 : $1 < t \le 2 \rightarrow \int_{t-1}^{t} 6\,d\tau = 6\tau\Big|_{t-1}^{t} = 6t - 6(t-1) = \boxed{6}$



Region 4 : $2 < t \le 3 \rightarrow \int_{t-1}^{2} 6\,d\tau = 6\tau\Big|_{t-1}^{2} = 12 - 6t + 6 =$ $\boxed{18-6t}$



Region 5: $t>3 \rightarrow \int_{-\infty}^{\infty} 0 \, d\tau = 0$



Discrete Convolution

$$X[n] = \sum_{\infty}^{\infty} x[k]\,\delta[n-k] \rightarrow y[n] = \sum_{-\infty}^{\infty} x[k] \cdot h[n-k]$$

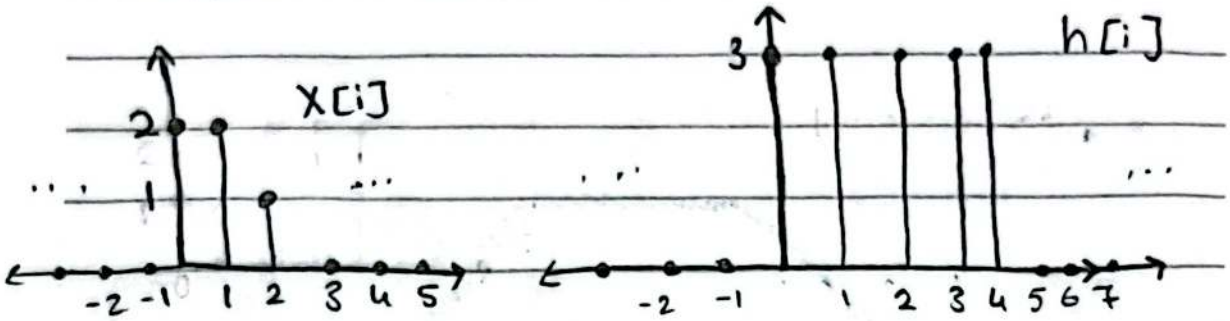
① if $h[n]=0$ for $n<0$ Thus $h[n-m]=0$ for $m>n$
Causal LTI System $\sum_{-\infty}^{n} x[m]\,h[n-m]$

② if $x[n]=0$ for $n<0$ soo Causal input & General LTI system $\sum_{0}^{\infty} x[m]\,h[n-m]$

③ if the system & signal are both causal then $\sum_{0}^{n} x[m]\,h[n-m]$

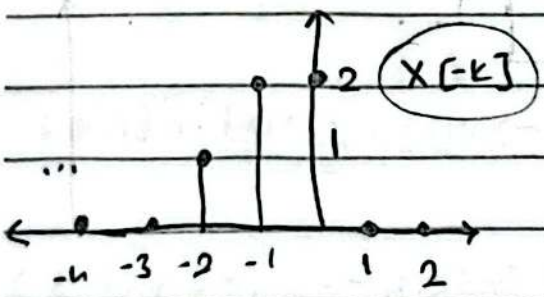
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Example :



Convolution

① $x[i] \rightarrow x[-k]$ (commutativity)



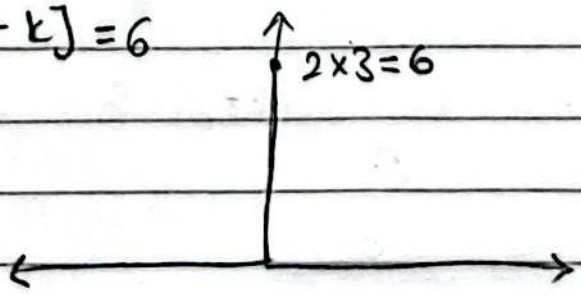
2 find Regions

1- for $n=0$ → shift → $x[0-k] = x[-k]$

$y[0] = h[k] \cdot x[-k] = 6$

$2 \times 3 = 6$

$\boxed{y[0] = 6}$



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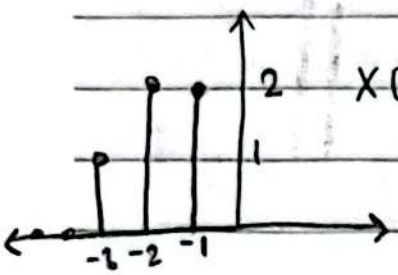
1 2 3 $x_1 \cdot y_3$

$x(1)\,y_1(1)$ $x(n)\,y_2$

$x_2 \cdot y_1$ $x_2 \cdot y_2$ $x_2 \cdot y_3$

2- for $n<0$ → left shift → $x[-1-k]$

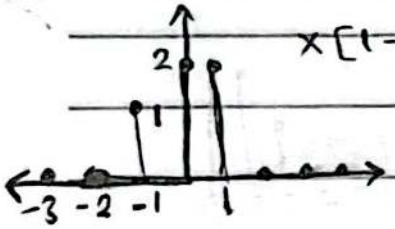
yani 1 adım Sola

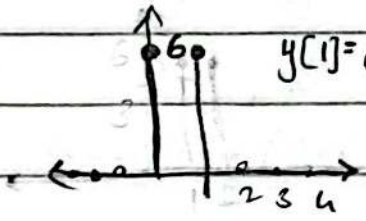


$y[n] = h[k] \cdot x[-1-k] = 0$

$\forall n<0$ çakışma yok

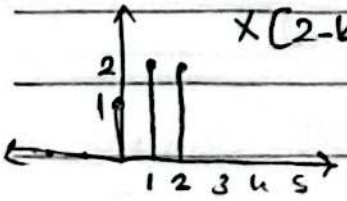
3- for $n=1$ → $x[1-k]$ 1 adım Sağa

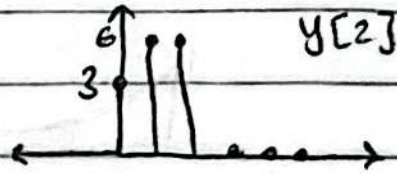




$y[1] = h[k] \cdot x[1-k] = 12$

4- for $n=2$ → 2 adım Sağa





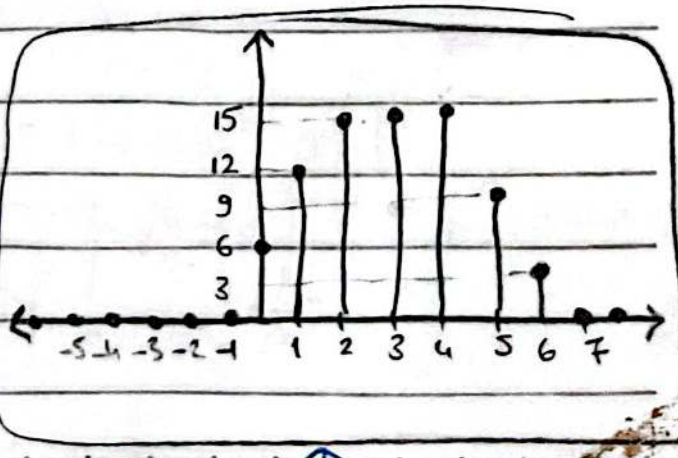
5- for $n=3 \rightarrow 15$

6- for $n=4 \rightarrow 15$

7- for $n=5 \rightarrow 9$

8- for $n=6 \rightarrow 3$

9- for $n>6 \rightarrow 0$



Solution

✳ Tabular Method:

$x_1[n] = \{1, 2, -2\}$ $\quad$ $x_2[n] = \{2, 0, 1\}$

$y[n] = x_1[n] * x_2[n] = ??$

	1	2	-2
2	2	4	-4
0	0	0	0
1	1	2	-2

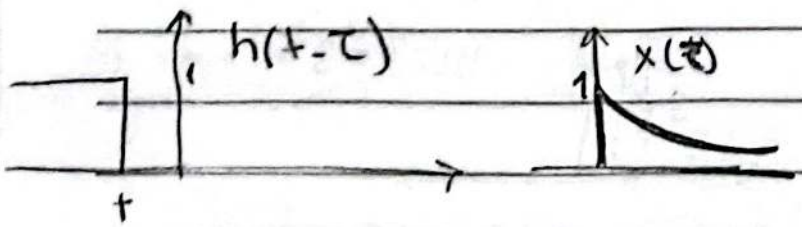
$y[n] = \{2, 4, -3, 2, -2\}$

$(2), (0+4), (1+0-4), (2+0), (-2)$

Örnek:

$x(t) = e^{-at} u(t)$, $a>0$ ve $h(t) = u(t)$ ise Sistemin çıkışını bulunuz.

$$y(t) = \int x(\tau)\, h(t-\tau)\, d\tau$$



$t=0 \rightarrow 0$

$t>0 \rightarrow \int_{-\infty}^{\infty} e^{-a\tau} u(t)\, u(t-\tau)\, d\tau \rightarrow \int_{0}^{t} e^{-a\tau}\, d\tau =$

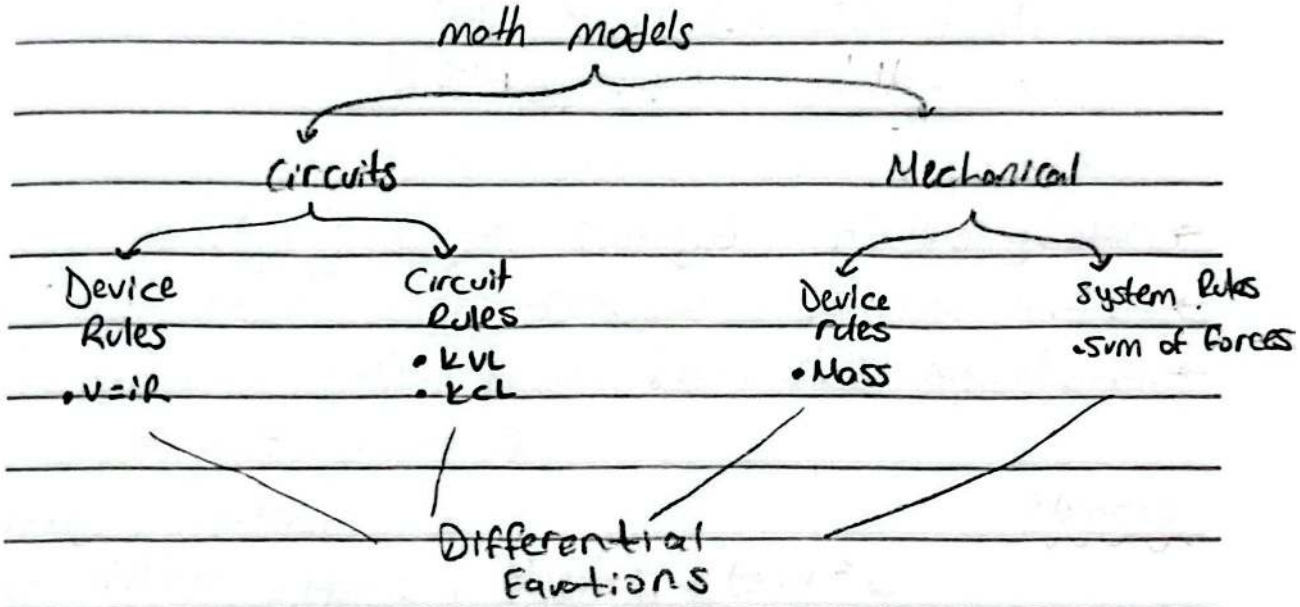
$$-\frac{1}{a} e^{-a\tau} \Big|_0^t = \frac{1}{a}(1 - e^{-at})$$

$$y(t) = \frac{1}{a}(1 - e^{-at}) \cdot u(t)$$

Sınav: Ders sonu: → Python matlab week dahil test + klas.k

Week - 7 - Transfer Function

System Modeling



Simple Model:

$$x(t) = V_R(t) + y(t)$$

$$V_R(t) = Ri(t)$$

$$i(t) = C\frac{dy(t)}{dt}$$

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t)$$

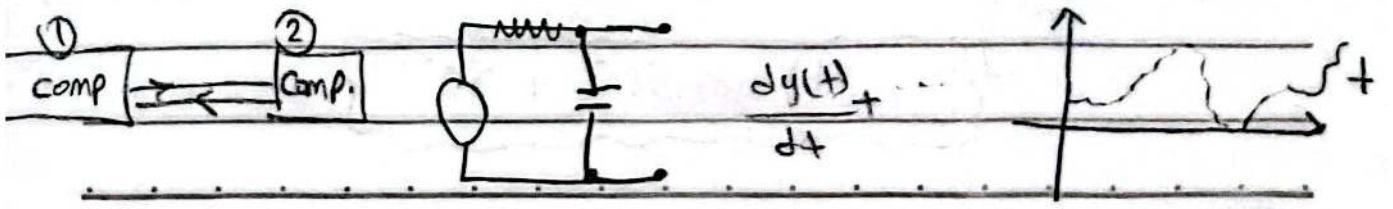
given : input $x(t)$ Find : Solution $y(t)$

1→ Sistemin eşitliğini bul

2→ dif denklem ise çözümünü bul

3→ Konuşsyon yap

Pystcal system → Shematic system → math sys. → math solution



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Differential Equations:

(Linear and time invariant)

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots$$

→ Ordinary Dif Eq : bağımsız değişkeni 1 tane

→ Partial dif Eq. : bağımsız en az 2.

→ homojen dif denklem → ... = 0

Example

$$\frac{d^2 y(t)}{dt} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{df(t)}{dt} \qquad y(0)=0 \quad y'(0)=-5$$

(*) $\frac{d}{dt} = D$ ile değiştir

$$D^2 y(t) + 3D y(t) + 2y(t) = 0$$

(*) $y(t)D = \lambda$

$$\lambda^2 + 3\lambda + 2 = 0 \rightarrow (\lambda+1)(\lambda+2)=0$$

karakteristik denklem

So $\lambda_1 = (-1) \quad \lambda_2 = (-2)$

$$y_{zi}(t) = C_1 e^{-t} + C_2 e^{-2t}$$

$$y(0) = C_1 + C_2 = 0$$

$$y'(0) = -C_1 e^{-0} - 2C_2 e^{-2.0} = C_1 + 2C_2 = 5$$

$$y_{zi}(t) = -5e^{-t} + 5e^{-2t}$$

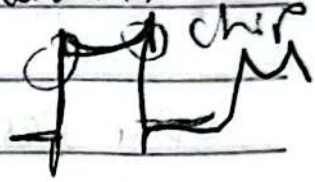
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hafta - 9 -

foto, frekans

Transfer functions

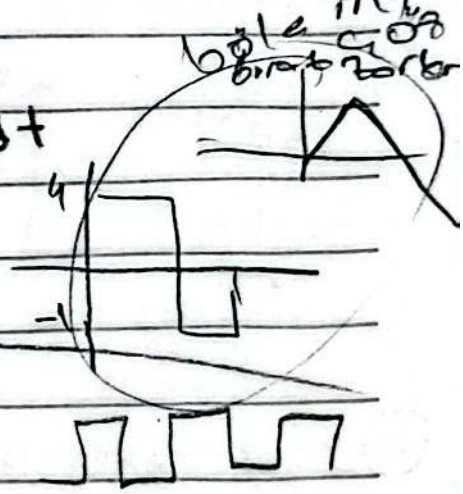
correlation: iki işaretin ne kadar benzediğini gösterir.
artı ∞ - ∞ benzer kaç frekans bulunur.



eğer işaret periodic ise buna fourier series integral uygulanabilir.

$$\frac{1}{T_0}\int x(t)\, e^{-j(2\pi k/T_0)t}\, dt$$

spectrum



analytically find FS coeff

① find T → then find w

② $c_k = \frac{\dot{a}}{2k\pi}\left(e^{-jk\pi} - 1\right) = \cos(k\pi) - j\sin(k\pi)$

$\sin(k\pi)$ ↓ hep sıfır

$\cos(k\pi) = 1, -1$

tek / çift

sin / cos

$a_0 = c_0$

$a_0/2 = R_0$

tree equivalent form of FS

exp → sin/0

$e^{j0{,}5\pi} = j$

$= \cos(0{,}5\pi) + j\sin(0{,}5\pi)$

$\frac{\sin(0{,}5\pi)}{1}$

$(A, w, \phi)(t?)$

faz genlik frekans (zaman?)

↓ kaybetti

pozitif olanın yeri → sonuç olur

$x(t) = 10 + 4($

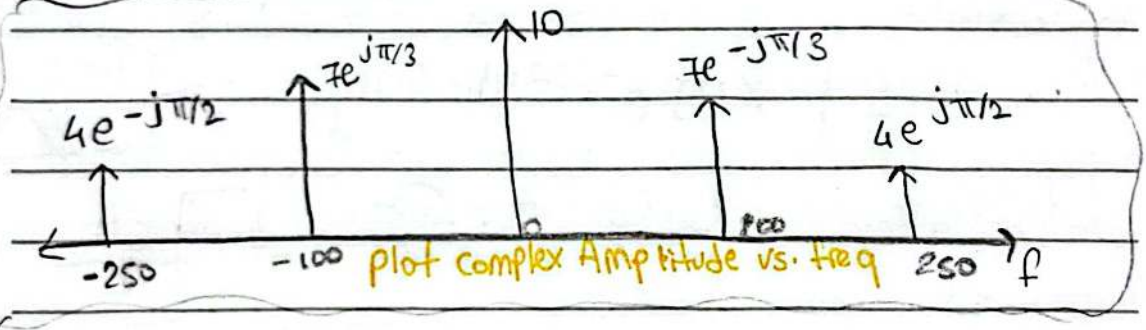
hafta -9- Fourier Series

fourier series Integral

$$a_k = \frac{1}{T_0}\int_0^{T_0} x(t)\, e^{-j(2\pi k/T_0)t}\, dt \qquad \left(e^{-j(\omega k)t}\right)$$

For periodic signals $\longrightarrow x(t+T_0) = x(t)$

a_k = complex Amplitude for kth harmonic



(*) euler's formula $\cos(\omega t) = 1/2\,(e^{j\omega t} + e^{-j\omega t})$

$\sin(\omega t) = 1/2j\,(e^{j\omega t} - e^{-j\omega t})$

hatırlatma $\boxed{-1/j = j = e^{j0.5\pi} = e^{j\pi/2}}$

hatırlatma-2:

$$A\cos(\omega t + \theta) = 1/2 A e^{-j\theta} e^{j\omega t} + 1/2 A e^{-j\theta} e^{-j\omega t}$$

(?) Convert Spectrum to Sinusoid: (*)

↳ Some graph:

freq	Amp	phase
-250	4	$-\pi/2$
-100	7	$\pi/3$
0	10	0
100	7	$-\pi/3$
250	4	$\pi/2$

$$x(t) = 10 + 4e^{j\pi/2} e^{j2\pi(250)t} + 4e^{-j\pi/2} e^{-j2\pi(250)t} + 7e^{j\pi/3} e^{j2\pi(100)t} + 7e^{-j\pi/3} e^{-j2\pi(100)t}$$

$$\hookrightarrow 10 + 8\cos(2\pi(250)t + \pi/2) + 14\cos(2\pi(100)t - \pi/3)$$

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$e^{j\omega t} = \cos(\omega t + \theta) + j\sin(\omega t + \theta)$

$a_k \quad b_k \quad \theta_k \quad T = f_0$

So we get the general form: $2\pi f = \omega$

$$X(t) = A_0 + \sum A_k \cos(\overbrace{2\pi f_k}^{\omega} t + \theta_k)$$

Fourier Series

allows us to write any real world periodic signal as a sum of sinusoids.

Yani periodik sinyali cos veya sin toplamına dönüştürür.

⊛ 3 forms of Fourier Series:

real-world ideas

① $X(t) = A_0 + \sum A_k \cos(k\omega_0 t + \phi_k)$ — Amplitude and phase form

doing math $c_k \Rightarrow$ phasor

② $X(t) = \sum c_k e^{jk\omega_0 t}$ — Complex exponential form

special scenarios

③ $X(t) = a_0 + \sum [a_k \cos(k\omega t) + b_k \sin(k\omega t)]$ — Sin Cos form

1-3 conversion

$A_0 = a_0$

$A_k = \sqrt{a_k^2 + b_k^2}$

$\theta_k = \tan^{-1}\left(\frac{-b_k}{a_k}\right)$

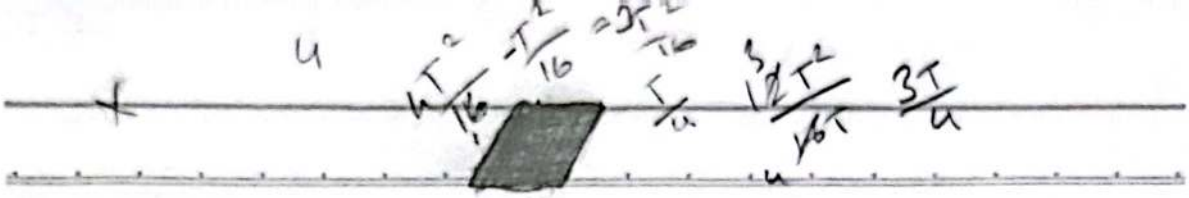
3-1 conversion

$a_k = A_k \cos(\theta_k)$

$b_k = -A_k \sin(\theta_k)$

$a_0 = c_0$

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1-2 conversion

$A_0 = c_0$

$A_k = 2|c_k|$ $k=1,2,3$

$\theta_k = \angle c_k$

2-1 conversion

$c_0 = A_0$

$c_k = 1/2\, A_k e^{j\theta_k}$

$c_{-k} = 1/2\, A_k e^{-j\theta_k}$

2-3 conversion

$c_0 = a_0$

$c_k = 1/2\,(a_k - jb_k)$

$c_{-k} = 1/2\,(a_k + jb_k)$

3-2 conversion

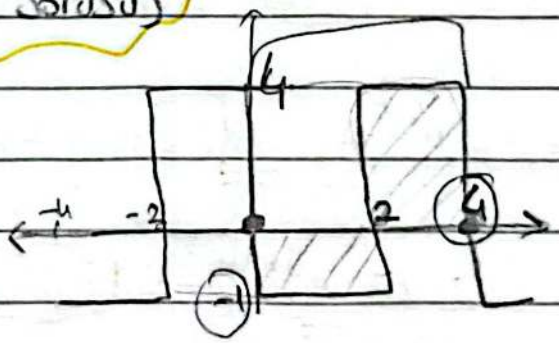
$a_0 = c_0$

$a_k = 2\,\text{Real}\{c_k\}$

$b_k = 2\,\text{imag}\{c_k\}$

* önce exponansiyeli bul en kolayı c_k

Soru (geçmiş Sınav Sorusu)



3)

① temel frekans:

$f_0 = gcd(f_1, f_2, f_3, \ldots)$

$T_0 = 4s \quad f_0 = 1/4 = 0{,}25$

② DC bileşen $= c_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt = 1/4 \cdot \int_0^2 -1\,dt + \int_2^4 4\,dt = 3/2$

1 periyot alanı

③ FS. $c_k = \frac{1}{T_0}\int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt}\,dt = \frac{1}{4}\int_0^2 -1\, e^{-j(2\pi/4)kt}\,dt + \frac{1}{4}\int_2^4 4\, e^{-j2\pi/4)kt}\,dt$

$c_k = \frac{1}{T}\int_{t_0}^{t_0+T} x(t) e^{-jk\omega t}\,dt = \frac{1}{4}\left[\frac{+e^{-j(\pi/2)kt}}{+j\frac{\pi}{2}k}\right]_0^2 + \left[\frac{4e^{-j\pi/2kt}}{-j\frac{\pi}{2}k}\right]_2^4$

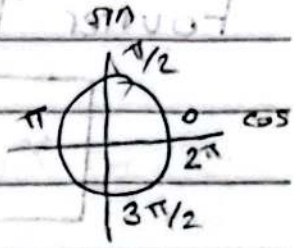
formül integral dönüşümü

$= \frac{j}{2\pi k} \times \left[-e^{-j\pi k} + 1\right] + \left(4e^{-j2\pi k} - 4e^{-j\pi k}\right)$

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4.30 - 5.30
5 saat

③ $J/2\pi k \times (1 + 4e^{-J2\pi k} - 5.e^{-J\pi k})$



$e^{-J2\pi k} = \underbrace{\cos(2\pi k)}_{(1)} - J\underbrace{\sin(2\pi k)}_{0}$

$e^{-J\pi k} = \underbrace{\cos(\pi k)}_{(-1)^k} - J\underbrace{\sin(\pi k)}_{0}$

$\frac{J}{2\pi k} \cdot \underbrace{\frac{1+4.1}{5}}_{} - 5.(-1)^k = \boxed{\frac{5J}{2\pi k} \cdot 1-(-1)^k = C_k}$

$C_1 = \frac{5J}{\pi}$ $\quad C_{-1} = -\frac{5J}{\pi}$

$C_2 = 0$ $\quad C_{-2} = 0$

$C_3 = \frac{5J}{3\pi}$ $\quad C_{-3} = -\frac{5J}{3\pi}$

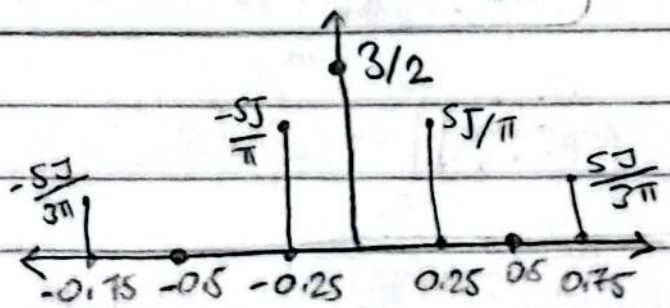
$a_k = 2(\text{Real } c_k)$

$b_k = 2(\text{imag } c_k)$

$a_k = 2.0 = 0$

$b_k = \frac{2.5(1-(-1)^k)}{2\pi k}$

$\boxed{b_k = \frac{5(1-(-1)^k}{\pi k}}$



⊛ eğer işaret periyodik ise Fourier Series kullanılır değilse Fourier Tranformation kullanılır

işareti $-\infty / +\infty$ arasında periyodik varsayılır

Laplace = $X(s) = \int_{-\infty}^{\infty} x(t) \cdot e^{-st} dt$

Fourier Transform

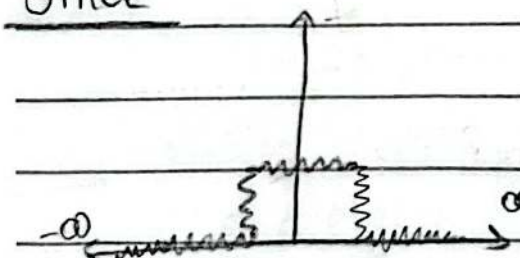
$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$

$X(w) \rightarrow w \in -\infty, \infty$

ters fourier tranform: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) e^{jwt} dw$

! yaptığımız şey zaman düzleminden frekans düzlemine geçiş yapmak $x(t) \longleftrightarrow X(jw)$ işaretin içine bakmak gibi içinde hangi sinüsoidaller var?

Örnek

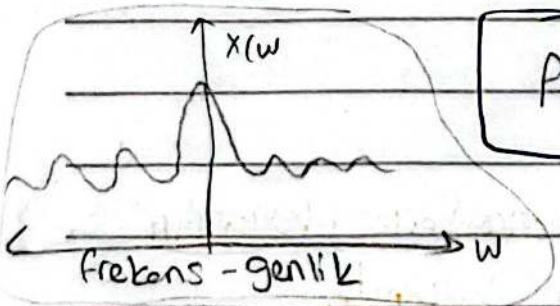


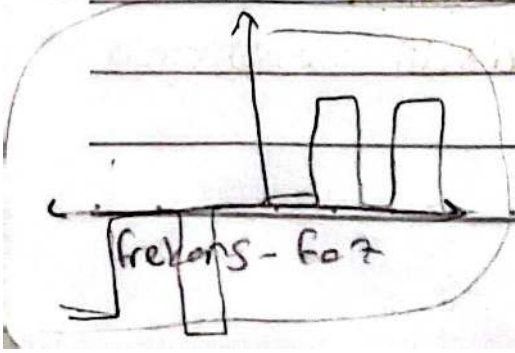
$$p_\tau(t) = \begin{cases} 1, & -\frac{\tau}{2} \leq t < \frac{\tau}{2} \\ 0, & \text{otherwise} \end{cases}$$

$$P_\tau(w) = \int_{-\infty}^{\infty} p_\tau(t)\, e^{-jwt} dt$$

$$\rightarrow \int_{-\tau/2}^{\tau/2} e^{-jwt} dt \rightarrow \frac{-1}{jw}\left[e^{-jwt}\right]_{-\tau/2}^{\tau/2} = \frac{2}{w}\left(\frac{e^{jw\frac{\tau}{2}} - e^{-jw\frac{\tau}{2}}}{2j}\right)$$

$$P_\tau(w) = \frac{2\sin\left(\frac{w\tau}{2}\right)}{w} \qquad \sin\left(\frac{w\tau}{2}\right)$$





test

$u(t) = \begin{cases} t \geq 0 , 1 \\ t < 0 , 0 \end{cases}$

Example-1-

$x(t) = e^{-at} u(t)$ (left sided) $\boxed{a>0}$

$$\int_{-\infty}^{\infty} e^{-\alpha t} \cdot u(t) \cdot e^{-j\omega t} dt = \int_{0}^{\infty} e^{-\alpha t} \cdot e^{-j\omega t} dt = \int_{0}^{\infty} e^{-(\alpha + j\omega)t} dt$$

$$\frac{e^{-(\alpha+j\omega)t}}{-(\alpha+j\omega)} \Big|_0^{\infty} = 0 - \frac{1}{-\alpha + j\omega} = 1/\alpha + j\omega$$

$$X(j\omega) = \frac{1}{a+j\omega}$$

Fourier Transform of $h(t)$

(*) $h(\omega)$: freqency response

not: $h(t) \rightarrow$ impulse response

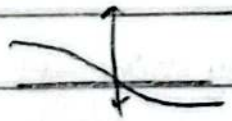
$x(t) * \delta(t) \rightarrow h(t)$

$$h(t) = e^{-t} u(t) \longleftrightarrow H(j\omega) = \frac{1}{1+j\omega}$$

$$|X(j\omega)| = 1/\sqrt{a^2+\omega^2}$$



$$\angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$



Example-2-

$$X(j\omega) = 2\pi\delta(\omega - \omega_0)$$

$$x(t) = 1/2\pi \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$x(t) = e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$x(t) = 1 \longleftrightarrow 2\pi\delta(\omega)$$

$$x(t) = \cos(\omega_0 t) \longleftrightarrow \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$ (stele)

$$\cos = \frac{e + c}{2}$$

Fourier transform tables

Decaying Exponential

$$x(t) = e^{-bt}u(t) \longleftrightarrow X(w) = \frac{1}{b+jw}$$

mg $\frac{1}{\sqrt{b^2+w^2}}$

ph $-\tan^{-1}\left(\frac{w}{b}\right)$

mg plot has even symetry

phase plot has odd symetry

$1 = 2\pi\delta(w)$

$u(t) = \pi\delta(w) + 1/jw$

$\delta(t) = 1$

$\cos(w_0 t) = \pi[\delta(w+w_0) + \delta(w-w_0)]$

$\sin(w_0 t) = j\pi[\delta(w+w_0) - \delta(w-w_0)]$

$e^{jw_0 t} = 2\pi\delta(w-w_0)$,

Properties

① Linearity: (Supremly important)

$x(t) \longleftrightarrow X(w) \quad y(t) \longleftrightarrow Y(w)$

$$\boxed{ax(t)+by(t)} \longleftrightarrow \boxed{aX(w)+bY(w}$$

② Time Shift: (Really imporft)

$x(t) \longleftrightarrow X(w)$ then $x(t-c) \longleftrightarrow X(w)e^{-jcw}$

if $c>0 \rightarrow$ delay

③ Time Scaling:

$a \neq 0$

$$x(at) \longleftrightarrow \frac{1}{|a|}X\left(\frac{w}{a}\right)$$

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④ Time Reversal:

$$x(-t) \longleftrightarrow X(-\omega)$$

⑤ Modulation Property:

uygun dönüşüm yapılması

$$x(t)e^{j\omega_0 t} \longleftrightarrow X(\omega-\omega_0)$$

⑥ Real sinusoid modulation:

shift down, shift

$$x(t)\cos(\omega_0 t) \longleftrightarrow \frac{1}{2}[X(\omega+\omega_0)+X(\omega-\omega_0)]$$

$$x(t)\sin(\omega_0 t) \longleftrightarrow 1/2\,[X(\omega+\omega_0)-X(\omega-\omega_0)]$$

⑦ Convolution:

$$y(t)=\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \longleftrightarrow Y(\omega)=X(\omega)\cdot H(\omega)$$

Sadece $X(\omega)$ ve $H(\omega)$ sonra $Y(\omega)$ için ters FT yap $y(t)$ ol

⑧ Parseval's theorem:

$$\int_{-\infty}^{\infty}|x(t)|^2dt=\frac{1}{2\pi}\int_{-\infty}^{\infty}|X(\omega)|^2d\omega$$

Convolution

$$Y(j\omega)=H(j\omega)\cdot X(j\omega)$$

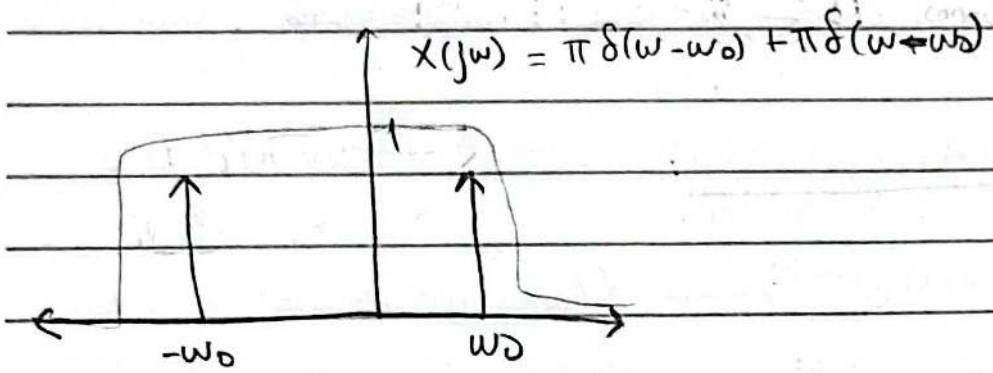
$$H(j\omega)\,[\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)]$$

$$y(t)=H(j\omega_0)\tfrac{1}{2}e^{j\omega_0 t}+H(-j\omega_0)\tfrac{1}{2}e^{-j\omega_0 t}$$

$$|H(j\omega_0)|\cos(\omega_0 t+\angle H(j\omega_0))$$

mag. faze

⑨ ideal Lowpass filter



gürültülü bir sesi filtrelemek istiyoruz varsayalım. Lowpass alttaki (negatif) olan frekansları yok edip üstteki(leri) kullanırız. highpass'te ise tam tersi

ideal → ya 1 ya 0

⑩ Duality:

$x(t) \rightarrow X(w) \quad X(w) \rightarrow x(t)$

Mutiply by t^n → diferans

Convolution → multiply

⑪ Diferentiation in the frequency Domain:

$y'(t) + 2y(t) = x(t) + x'(t) \rightarrow jwY(w) + 2Y(w) = X(w) + jwX(w)$

$(jw+2)Y(w) = (1+jw)X(w)$

$H(w) = \frac{Y(w)}{X(w)} = \frac{1+jw}{2+jw} = 1 - \frac{1}{2+jw} \rightarrow$ F+-1

$h(t) = \delta(t) - e^{-2t}u(t)$

Dirichlet conditions (fourier almanın koşulları)

① the func must be absolutely integrable

② Bounded variation (BIBO)

③ finite number of discontinuities (sonsuz süreksizlik)

fourier alamadığımız durumlarda → Laplace transform

$$\text{(X)}\quad L(x(t)) = X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$x(t) \overset{L}{\longleftrightarrow} X(s)$$

→ yakınsamay fonksiyonu yakınsak hale getirip Laplace dönüşümü alınır

$$A\int_0^{\infty} e^{-jwt}\, dt \longrightarrow A\int_0^{\infty} e^{-(\sigma + jw)t}\, dt$$

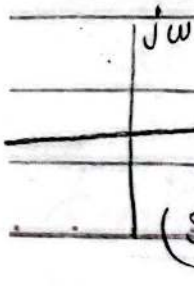
(doesn't converge; real part; converge (if $\sigma > 0$))

Örnek: $x(t) = e^{-at}\, u(t)$ → Laplace?

$$\int_0^{\infty} e^{-at} \cdot u(t) \cdot e^{-st}\, dt \rightarrow \int_0^{\infty} e^{-(s+a)t}\, dt$$

$$\frac{-e^{-(s+a)t}}{s+a}\Bigg|_0^{\infty} = \frac{-e^{-(s+a)\cdot\infty}}{s+a} - \left(\frac{-e^{-(s+a)\cdot 0}}{s+a}\right) =$$

$$s = \sigma + jw \qquad \frac{e^{\infty}}{s+a} = 0 - \frac{-1}{s+a} \qquad \frac{-1}{s+a}$$



$$= \frac{1}{s+a}$$

$(s+a) > 0$ olmalı

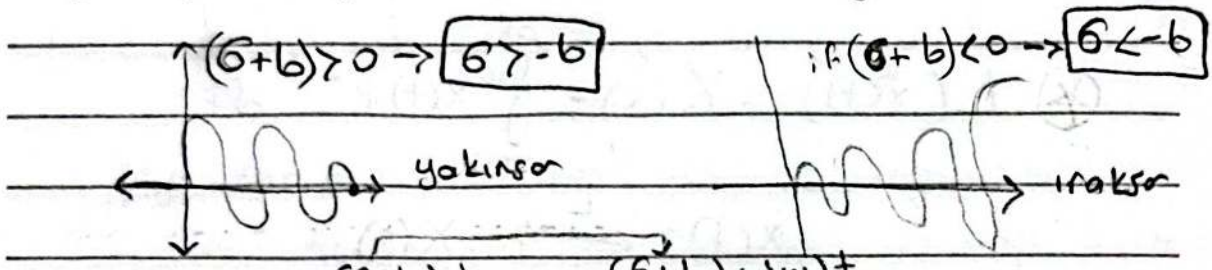
$\sigma + jw + a > 0 \rightarrow \sigma > -a$ olursa

2 types of laplace → 2 sided (bilateral) → $\int_{-\infty}^{\infty}$
→ 1 sided (unilateral) → $\int_{0}^{\infty}$ or $\int_{-\infty}^{0}$

Region of convergence (ROC)

each signal has its own region of convergence

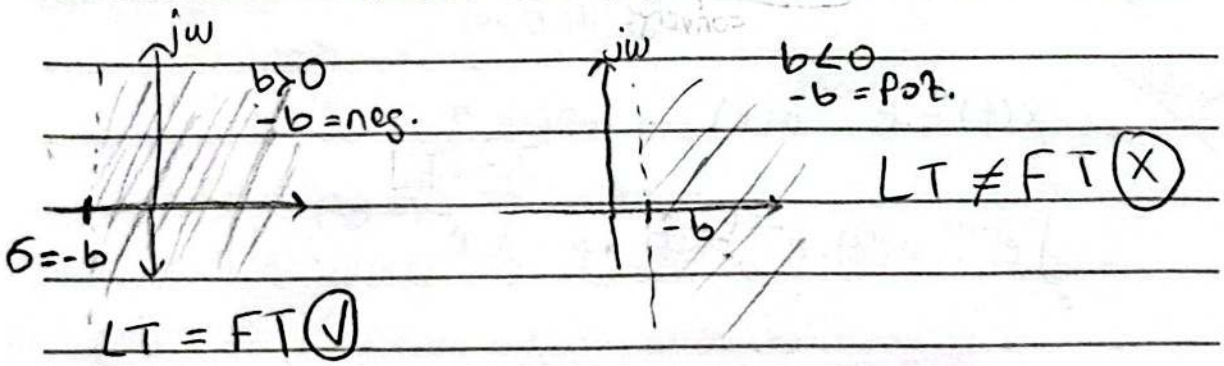
Sağa yakınsayabilir sola ıraksar ya da tersi gibi



$e^{-(s+b)t} = e^{-(\sigma+b)+j\omega)t}$

so $\sigma > -b$ $X(s) = 1/s+b$ and $Re\{s\} > -b$

⊛ if ROC includes the jw axis then FT=LT



$e^{-at}u(t) \xleftrightarrow{L} 1/s+a$, $\sigma > -a$ Right sided

$-e^{-at}u(-t) \xleftrightarrow{L} 1/s+a$, $\sigma < -a$ Left sided

⊛ Laplace dönüşümlerine tekrar bak (Diferansiyel)

example

Let $x(t)=u(t)$ and $h(t)=e^{-4t}u(t)$ find $y(t)$?

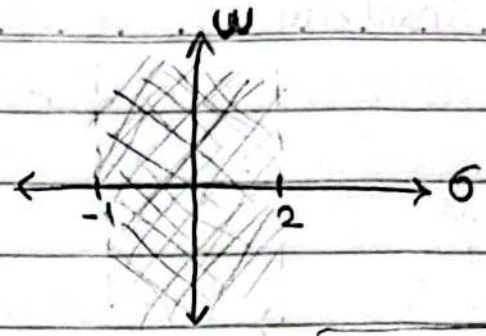
$x(t)$ → $1/s$; $h(t)$ → $\frac{1}{4+s}$; $y(t)$ → $\frac{1}{4}\cdot 1-e^{-4t}\cdot u(t)$

$y(t) = x(t) * h(t) \rightarrow \frac{1}{s}\cdot\frac{1}{4+s} = \frac{1}{s^2+4s} = \frac{1/4}{s} - \frac{1/4}{s+4}$

(*) σ ile ilgili bilgi vermezse u(t) kabul et

example-2-

Find the laplace tranform of $x(t) = e^{-t}u(t) + e^{2t}u(-t)$



$e^{-t}u(t) = 1/(s+1)$ $\sigma > -1$

$e^{2t}u(-1) = -1/(s-2)$ $\sigma < 2$

$x(t) \longleftrightarrow (1/(s+1)) - (1/(s-2)),\ -1 < \sigma < 2$ **LT = FT**

Properties of Bilateral Laplace Transform (2-sided) time shi

① Linearity : $ax(t) + by(t) = aX(s) + bY(s)$ ② $x(t-c) \longleftrightarrow e^{-cs}X(s)$

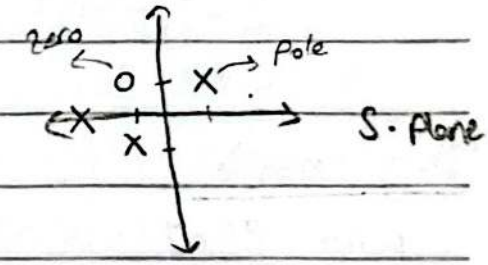
③ $\dot{x}(t) \longleftrightarrow sX(s)$ (türev) ④ $\int_{-\infty}^{t} x(m)\,dm \longleftrightarrow 1/s\,X(s)$ (bölü)

⑤ $X(s) \cdot H(s) = Y(s)$ convolution

(*) hatırlatma: $\mathcal{L}\{y'(t)\} = s \cdot Y(s) - y(0)$

Poles and Zeros

$$H(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$$ → zeros, → poles



Payı sıfır yapan = zeros (sıfır) paydayı sıfır yapan = poles (sonsuz)

Discrete time fourier Transform (*)

$x[n] = 1/2\pi \int_{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega \rightarrow X(\Omega) = \sum x[n] e^{-j\Omega n}$

mesela $x[n] = [1\ 2] \rightarrow x[0] \cdot e^{-j\Omega 0} + x[1] \cdot e^{-j\Omega 1} = X(\Omega)$

CTFT → laplace DTFT → Z transform

↓ disc. time fourier

kullanılır

Z tranform can be applied to .....

FIR filter : $h[n] \rightarrow H[z]$

Signals : $x[n] \rightarrow X[z]$ $\quad H[z] = \sum_n h[n] z^{-n}$

$x[n] \longrightarrow h[n] = \delta[n] - \delta[n-1] \longrightarrow y[n]$

$H(e^{j\hat{w}}) = \sum h[n] e^{-j\hat{w}n}$

$H(e^{j\hat{w}}) = 1 \cdot e^{0} - 1 e^{-j\hat{w}} = \boxed{1 - e^{-j\hat{w}}}$

(*) $\boxed{H(z) = \sum_n h[n] z^{-n}}$ (Polynomial representation)

(example)

$h[n] = \{2, 0, -3, 0, 2\}$

$H[z] = 2 \cdot z^{0} + 0 z^{-1} - 3 z^{-2} + 0 z^{-3} + 2 z^{-4}$

$= 2 - 3(z^{-1})^2 + 2(z^{-1})^4 \Rightarrow 2 - 3x^2 + 2x^4$ gibi

Delay example

Find $y[n]$ via polynomials : $X[n] = 3, 1, 4, 1, 5, 9, 0, 0, \ldots$

$Y(z) = z^{-1} X(z) = z^{-1} \cdot (3 + z^{-1} + 4z^{-2} + z^{-3} + 5z^{-4} + 9z^{-5})$

$Y(z) = 0 z^{0} + 3 z^{-1} + z^{-2} + 4 z^{-3} + z^{-4} + 5 z^{-5} + 9 z^{-6}$

(*) for -1 delay multiply by z^{-1} (or for n delay $\times z^{-n}$)

ALIASING Derivation

(*) cos inişin açısının ibi katıyla eklenmesi